Call Manager (Aegis Calling) — Requirements Specification

Project: AI-powered outbound admissions calling platform **Client:** The Shri Ram Academy (TSRA), Gachibowli, Hyderabad **Document version:** 1.0 **Status:** Live in production (AWS deployment)

---

1. Purpose

Automate admissions outreach calling for TSRA: an AI voice agent calls prospective families, answers questions about the school, books campus visits, schedules callbacks, and syncs results to Google Calendar and email — without a human dialer for first-contact outreach.

---

2. Stakeholders / User Roles

Role	Access	Responsibilities
Admin	Full dashboard access	Upload leads, manage campaigns, configure Settings (API keys, SMTP, dialer limits), view all data
Staff	Dashboard access, no Settings	View/manage contacts, campaigns, appointments, callbacks
Prospective family (caller)	Phone only	Receives/answers calls from the AI agent

---

3. Functional Requirements

FR-1 — Campaign Management

- FR-1.1: Admin can upload a lead list (Excel/CSV) to create a campaign ("UploadBatch").
- FR-1.2: System auto-dials contacts in a campaign up to a configurable concurrency limit.
- FR-1.3: Each campaign has an independent dial queue (one campaign's pace never blocks another).
- FR-1.4: As a call ends, the next queued contact in that campaign is dialed automatically.
- FR-1.5: Do-Not-Call contacts (by phone number) are excluded from any future dial, campaign or otherwise.

FR-2 — AI Voice Conversation

- FR-2.1: The agent ("Maya") opens every call with a fixed identity check.
- FR-2.2: The agent answers factual questions about the school using only

grounded, live-scraped knowledge — **never invented facts**.

- FR-2.3: The agent converses in **English, Hindi, or Tamil**, matching the

caller's language, including natural code-switching ("Hinglish").

- Telugu is explicitly **out of scope** — no provider on the underlying voice platform supports Telugu STT or TTS. The agent must say so plainly rather than guess.
- FR-2.4: The agent asks for one piece of information at a time during booking (date/time, then purpose, then email) — never bundles multiple open-ended questions into a single turn.
- FR-2.5: The agent must never resolve a date, time, purpose, or email from an ambiguous reply (e.g. a bare "yes" to a multi-part question) — it must ask again for the specific missing piece.
- FR-2.6: The agent recognizes colloquial/multilingual affirmatives ("ha", "haan", "achha", "aama", etc.) as "yes", including short/garbled variants.
- FR-2.7: A spoken promise (booking confirmed, callback scheduled) is only valid if the corresponding backend tool was actually invoked and returned success — the agent must never narrate an outcome it didn't actually produce.

FR-3 — Reschedule / Callback Flow

- FR-3.1: Caller can request to be called back at a specific time; the agent resolves relative time expressions ("in 10 minutes," "tomorrow evening") against the actual call timestamp (IST).
- FR-3.2: A callback is created via a real tool call and confirmed to the caller only after that tool call succeeds.
- FR-3.3: If a scheduled callback's target time arrives while the contact is **already on an active call** (from any trigger — another callback, a manual call, or a campaign dial), the callback must defer rather than double-dial the contact.
- FR-3.4: If a call goes unanswered, or ends with nothing resolved, the system automatically retries — capped at 2 automatic retries (3 total attempts) before flagging the contact for manual follow-up.

FR-4 — Appointment Booking

- FR-4.1: Caller can book a campus visit / admission counseling session; the agent collects and confirms date, time, purpose, and email before booking.
- FR-4.2: Booking a second time for a contact with an existing "Booked" appointment updates the existing record in place — never creates a duplicate.
- FR-4.3: On successful booking, a Google Calendar event is created and a confirmation email is sent — both as background tasks with **zero added latency** to the live call.
- FR-4.4: If the caller states the on-file email is incorrect, the agent must obtain and confirm a corrected email before booking — never book with an empty or unconfirmed email address.

- FR-4.5: If a tool call fails, the agent must not claim success; it must

reassure the caller their request is noted and schedule a callback so the request isn't lost.

FR-5 — Knowledge Base

- FR-5.1: The school's website is scraped periodically (and on-demand) into

searchable knowledge chunks.

- FR-5.2: Every factual answer must be grounded in a real, current lookup —

the agent must actually invoke the lookup tool, never skip straight to an "I don't have that information" apology.

- FR-5.3: If a lookup genuinely returns nothing relevant, the agent offers

to have the admissions team follow up rather than guessing.

FR-6 — Dashboard

- FR-6.1: Campaigns page — upload leads, view/start/monitor campaigns.
- FR-6.2: Contacts/Leads page — per-contact status, call history, transcripts, recordings.
- FR-6.3: Scheduling page — view/manage scheduled callbacks and booked appointments.
- FR-6.4: Settings page (admin only) — configure Retell, Google Calendar, SMTP, dialer limits, tool auth secret.
- FR-6.5: JWT-based session auth with role separation (admin vs staff).

FR-7 — Outcome Tracking

- FR-7.1: Every call ends with exactly one recorded outcome: `interested_followup_scheduled`, `appointment_booked`, `not_interested`, `do_not_call`, `wrong_number`, `no_answer`, or `undetermined`.
- FR-7.2: Outcome recording drives contact status and the dashboard's reporting.

---

4. Non-Functional Requirements

NFR-1 — Reliability

- NFR-1.1: The shared voice agent's configuration (prompt, tools, webhook)

must self-configure on backend startup, on any deployment, without a manual setup step. Agent/LLM identity must be persisted in the shared database (not a local file) so it survives across deployments.

- NFR-1.2: Backend tool endpoints (booking, scheduling, lookup) must respond

within the voice platform's tool-call timeout under normal conditions; database connections must survive idle periods without silently hanging (TCP keepalives + connection pre-validation + periodic recycling).

- NFR-1.3: A contact must never be dialed twice simultaneously by two

different triggers (scheduled callback, manual call, campaign dial).

NFR-2 — Security

- NFR-2.1: Tool webhook endpoints should be protected by a shared secret

header, verified on every request, once configured.

- NFR-2.2: Session auth (JWT) must use a non-default, random secret in production.
- NFR-2.3: Admin/staff credentials must be rotated from demo defaults before handling real lead data.
- NFR-2.4: Webhook payloads from the voice platform should be signature-verified when a real API key is configured.

NFR-3 — Performance

- NFR-3.1: Booking/scheduling must not add latency to the live call — any

slow operation (calendar sync, email) runs as a background task.

- NFR-3.2: Backend and database should be co-located in the same

geographic region as end users to minimize dashboard/API latency.

NFR-4 — Data Integrity

- NFR-4.1: Contact records must be resolvable unambiguously from a live

call (via call ID, then phone, then email) — duplicate contacts sharing a phone number must not cause bookings/callbacks to attach to the wrong record.

- NFR-4.2: The database used by this project must be logically isolated

from unrelated applications' data.

NFR-5 — Observability

- NFR-5.1: Every webhook event and tool call should be logged with enough

context (call ID, contact ID) to trace a live incident after the fact.

---

5. System Architecture

```
Frontend (React 19 + TS)  --HTTPS-->  Backend (FastAPI)  -->  PostgreSQL (dedicated DB)
                                               |
        +----------------+----------------+----------------+------------------+
        v                v                v                v                  v
   Voice AI         Google           SMTP/Gmail      TSRA Website     Shared voice
   (Retell -        Calendar         (confirmation    (scraped for     agent config
   calls, STT/TTS,  (service         emails)          knowledge base)  (self-configured
   webhooks)        account)                                           on startup)
```

- **Frontend**: React 19 + TypeScript + Vite. 4 pages: Campaigns, Leads, Scheduling, Settings.
- **Backend**: FastAPI + SQLAlchemy. REST API + APScheduler background jobs + webhook receivers.
- **Voice agent**: One shared Retell agent ("Maya," voice `11labs-Monika`), driven entirely by `agent_prompt.md` — no per-campaign forks.

- **Database**: Dedicated PostgreSQL instance — 7 tables (see §6).

---

6. Data Model (7 core tables)

Table	Purpose
`contacts`	Leads — name, phone, email, status
`upload_batches`	Campaigns (one per uploaded file)
`call_attempts`	Every call — outcome, duration, transcript, recording URL
`scheduled_callbacks`	Pending/fired reschedule requests
`appointments`	Booked campus visits, with Google Calendar links
`knowledge_chunks`	Scraped website content for factual answers
`settings`	Runtime-configurable credentials/config, including the shared agent's persisted `local_agent_id` / `retell_llm_id`

---

7. External Integrations

System	Purpose	Notes
Retell AI	Voice calls, STT/TTS, LLM orchestration	Single shared agent across all campaigns
Google Calendar API	Appointment events	Service account; cannot add formal "attendees" without Workspace domain delegation — caller notified via email instead
Gmail SMTP	Booking confirmation emails	Requires an App Password, not the account password
TSRA website	Source of truth for factual answers	Scraped nightly + on-demand refresh

---

8. Known Constraints / Out of Scope

- **Telugu is not supported** by the underlying voice platform for either

speech recognition or speech synthesis, on any provider. This is a platform-level limitation, not a configuration gap — no setting can add it.

- **Google Calendar attendee invites** are not possible with the current

service-account setup (requires a paid Google Workspace domain with delegated authority).

- Caller-initiated reschedule requests ("call me back later") are not capped — a caller can request unlimited callbacks; only the system-initiated no-answer retry is capped at 3 total attempts.

- The voice platform's "multi" legacy language mode is intentionally not

used — it covers a wider but less accurate set of languages (including documented cross-language misdetection); the system uses an explicit English/Hindi/Tamil locale set instead for better accuracy.

---

9. Acceptance Criteria Summary

A call is considered correctly handled when:

- No factual claim is made without a grounded knowledge lookup.
- No booking/callback detail (date, time, purpose, email) is invented —

every value traces back to something the caller actually said.

- Every spoken confirmation of a booking or callback corresponds to an

actual, successful backend tool call.

- The contact is never dialed by two different triggers at once.
- The outcome recorded matches what actually happened on the call.

This document reflects requirements established and validated through extensive live-call testing during initial development and the AWS production migration.